

You may use an LLM of your choice (ChatGTP, Claude, Gemini, etc) as well as your own non-artificial intelligence to answer questions 1 and 2 in this assignment. If you do, make sure your LLM gives you correct answers - they often make mistakes.

1. This problem is based on the duopoly model of trade developed in Lecture 8. Use the following values for this question: $a = 30, b = 4, d = 2$. In addition, suppose that marginal cost for the firms in the two countries are $MC_1 = 2$ and $MC_2 = 6$.
 - (a) Calculate autarky equilibrium price, quantity, profit, and consumer welfare (defined as utility minus expenditure) for both countries. What accounts for the differences, if any, between the two countries?
 - (b) Assume each country's firm can now sell in the other country's market. What is each firm's reaction function?
 - (c) For each of the two countries, calculate the price and quantity of both goods consumed (that is, the domestically produced good and the imported good). Also calculate total profit (including profits earned from both domestic and export sales) and consumer welfare for both countries.
 - (d) What is the intraindustry trade (IIT) index?
 - (e) Do consumers and producers in both countries gain from trade? Explain.
2. The data source for this problem is Exhibits 7 "U.S. Exports of Goods by End-Use Category and Commodity" and 8 "U.S. Imports of Goods by End-Use Category and Commodity" from the PDF included with this assignment.
 - (a) For 2024, extract data for all goods that had a value of trade (imports plus exports) of at least \$1 billion. For these goods, create a table that shows imports, exports, and the intraindustry trade (IIT) index. Drop any goods that have either zero imports or zero exports.
 - (b) What is the average value of IIT in the table you just created?
 - (c) From your table created in part (a), create a table showing the 5 goods with the highest and the 5 goods with the lowest levels of IIT. Include the levels of imports, exports, and IIT in your table.
 - (d) Of the 10 goods you analyzed in part (c), do any surprise you by their high or low levels of IIT? Explain.

Q1

$$a = 30, \quad b = 4, \quad d = 2, \quad MC_1 = 2, \quad MC_2 = 6$$

$$U(Q_1, Q_2) = a(Q_1 + Q_2) - \frac{b}{2}(Q_1^2 + Q_2^2) - d Q_1 Q_2$$

Demand function :

$$P_1 = a - bQ_1 - dQ_2, \quad P_2 = a - bQ_2 - dQ_1$$

(a) Autarky

Each firm i faces the demand function : $P_i = a - bQ_i$

$$\text{Profit is } \pi_i = \max_{Q_i} \{ (a - bQ_i - MC_i) Q_i \}$$

$$\text{FOC : } \underbrace{a - 2bQ_i}_{\text{autarky } MR_i} - MC_i = 0$$

$$\Rightarrow \overset{\text{autarky}}{Q_i^A} = \frac{a - MC_i}{2b}, \quad P_i^A = \frac{a + MC_i}{2}$$

→ Country 1 (Firm 1)

$$Q_1^A = \frac{30 - 2}{8} = 3.5, \quad P_1^A = \frac{30 + 2}{2} = 16$$

$$\text{Profit : } \pi_1^A = (16 - 2) \times 3.5 = 49$$

$$\text{Utility : } U_1^A = 30 \times 3.5 - 2 \times 3.5^2 = 80.5$$

$$\text{Expenditure : } P_1^A Q_1^A = 16 \times 3.5 = 56$$

$$\text{Consumer welfare : } CW_1^A = 80.5 - 56 = \boxed{24.5}$$

→ Country 2 (Firm 2)

$$Q_2^A = \frac{30 - 6}{8} = 3, \quad P_2^A = \frac{30 + 6}{2} = 18$$

$$\text{Profit : } \pi_2^A = (18 - 6) \times 3 = 36$$

$$\text{Utility : } U_2^A = 30 \times 3 - 2 \times 3^2 = 72$$

$$\text{Expenditure : } P_2^A Q_2^A = 18 \times 3 = 54$$

$$\text{Consumer welfare : } CW_2^A = 72 - 54 = \boxed{18}$$

(b) Trade

→ Firm 1

$$\text{Demand : } P_1 = 30 - 4Q_1 - 2Q_2$$

$$MR_1 = 30 - 8Q_1 - 2Q_2$$

$$\text{Profit maximization condition : } MR_1 = MC_1$$

$$\Rightarrow 30 - 8Q_1 - 2Q_2 = 2$$

$$\Rightarrow \boxed{Q_1 = 3.5 - \frac{1}{4}Q_2}$$

→ Firm 2

$$\text{Demand : } P_2 = 30 - 4Q_2 - 2Q_1$$

$$MR_2 = 30 - 8Q_2 - 2Q_1$$

$$\text{Profit maximization condition : } MR_2 = MC_2$$

$$\Rightarrow 30 - 8Q_2 - 2Q_1 = 6$$

$$\Rightarrow \boxed{Q_2 = 3 - \frac{1}{4}Q_1}$$

(c). Trade Equilibrium

Demand Country 1

$$\text{Firm 1} \quad P_1^1 = 30 - 4Q_1^1 - 2Q_2^1$$

$$\text{Firm 2} \quad P_2^1 = 30 - 4Q_2^1 - 2Q_1^1$$

⇓

Reaction function Country 1

$$\text{Firm 1} \quad Q_1^1 = 3.5 - \frac{1}{4}Q_2^1$$

$$\text{Firm 2} \quad Q_2^1 = 3 - \frac{1}{4}Q_1^1$$

⇓

$$Q_1^1 = \frac{44}{15} \approx 2.933$$

$$Q_2^1 = \frac{34}{15} \approx 2.267$$

Country 2

$$P_1^2 = 30 - 4Q_1^2 - 2Q_2^2$$

$$P_2^2 = 30 - 4Q_2^2 - 2Q_1^2$$

Country 2

$$Q_1^2 = 3.5 - \frac{1}{4}Q_2^2$$

$$Q_2^2 = 3 - \frac{1}{4}Q_1^2$$

⇓

$$Q_1^2 = \frac{44}{15} \approx 2.933$$

$$Q_2^2 = \frac{34}{15} \approx 2.267$$

$$\Rightarrow \begin{aligned} P_1^1 &= P_1^2 = \frac{206}{15} \approx 13.733 \\ P_2^1 &= P_2^2 = \frac{226}{15} \approx 15.067 \end{aligned}$$

$$\text{Profit: } \pi_i = \sum_j (P_i^j - MC_i) Q_i^j$$

$$\pi_1 = 68.836$$

$$\pi_2 = 41.102$$

$$\text{Utility: } U^1 = U^2 = 115.218$$

$$\text{Expenditure: } P_1^j Q_1^j + P_2^j Q_2^j = 14.436 \quad \text{for } j = 1, 2$$

$$\text{Consumer welfare: } U^j - P_1^j Q_1^j - P_2^j Q_2^j = 40.782 \quad \text{for } j = 1, 2$$

(d). Intra industry trade index

$$IIT = 1 - \frac{|X - M|}{X + M}$$

$$X_1 = P_1^2 Q_1^2 = M_2 = 13.733 \times 2.933 = 40.284$$

$$M_1 = P_2^1 Q_2^1 = X_2 = 15.067 \times 2.267 = 34.151$$

$$\Rightarrow IIT = 0.918$$

(e). Gain from Trade

		Autarky	Trade	GFT
Country 1	Consumer welfare	24.5	40.782	+ 16.282
	Producer profit	49	68.836	+ 19.836
Country 2	Consumer welfare	18	40.782	+ 22.782
	Producer profit	36	41.102	+ 5.102